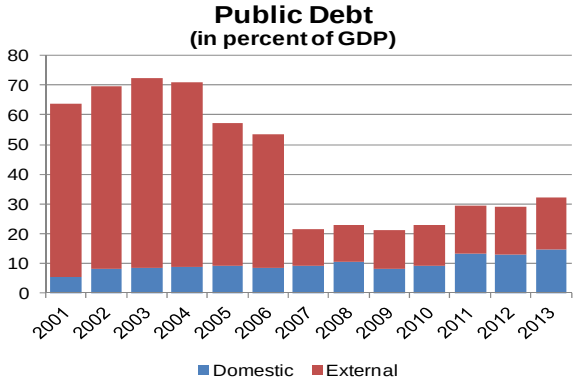
# Annex 1. Debt Sustainability Analysis Update

*This debt sustainability analysis (DSA) updates the joint IMF/IDA DSA from June 17, 2013.[1] It incorporates recent macroeconomic developments and the planned scaling up of non-concessional borrowing (NCB) from $1.5 billion to $2.2 billion to finance critical infrastructure projects. The DSA also incorporates contingent liabilities arising from two public-private partnership (PPP) projects. In an alternative scenario, the DSA takes account of a potential additional NCB increase to finance the Ugandan share of a regional railway project aimed at advancing EAC integration. Another scenario illustrates the impact of oil production set to start in 2018. Results indicate that Uganda remains at a low risk of debt distress. Nonetheless, the debt service-to-revenue ratio is high owing to the relatively short maturity of domestic debt and poses some risks.*

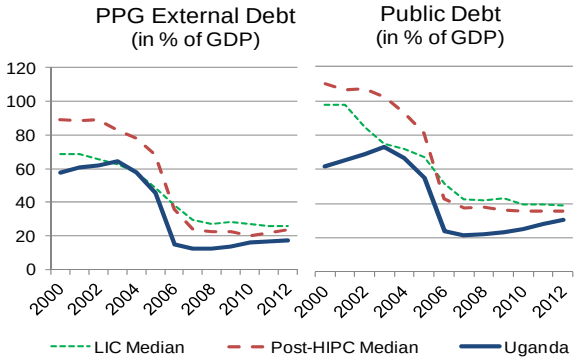
## A. Background and Recent Developments

**1. Since Uganda benefited from debt relief, the authorities have been cautious in accumulating new external debt while steadily developing the domestic debt market.** At about 5 percent of GDP, the increase in external debt in 2007–2013 was limited. In the same period, public domestic debt increased by 5½ percent of GDP, reflecting efforts to diversify funding sources and develop the market. The composition of public debt has been fairly stable, with the share of external debt remaining in the range of 55–60 percent.




Source: The authorities and IMF staff.
Note: Data provided by fiscal year covering July to June.

**2. The external debt burden compares favorably to other post-MDRI countries.** Uganda's external debt to GDP ratio dropped to 12¼ percent of GDP in 2007, becoming one of the lowest among recipients of Multilateral Debt Relief Initiative (MDRI) relief (text chart). Since then, both external and total public debt- to-GDP ratios have remained substantially below both the LIC and post-HIPC countries medians. The gap with both has narrowed as more countries received MDRI relief; while early beneficiaries, including Uganda, accumulated debt. Uganda's external debt stock increased by only about 4¾ percent of GDP in 2007–2012 compared to a median increase of 6 percent of




[1] In line with the 2010 Staff Guidance Note, a full joint LIC DSA is expected to be prepared once every three years for PRGT-eligible IDA-only countries. In between, short annual updates are produced unless macroeconomic conditions since the last full DSA have significantly changed. See Staff Guidance Note on the Application of the Joint Fund-Bank Debt Sustainability Framework for Low-Income Countries (www.imf.org).

GDP for early beneficiaries of MDRI. The cautious approach in contracting new external debt maintains Uganda's external debt in the lowest quartile within the countries that qualified for MDRI in 2006 (Text Table 1). New debt has mainly financed infrastructure projects and enhancement of transparency and service delivery in the public sector.

**3. Nonetheless, domestic borrowing in Uganda increased faster than in its peer countries.** Between 2007 and 2012 Uganda's domestic debt stock picked up from 9 percent to 13.1 percent of GDP—higher than about two-thirds of early beneficiaries of MDRI. Nevertheless, at 30¼ percent of GDP at end-2012, Uganda's total public debt is still lower than the median for the early MDRI beneficiaries.

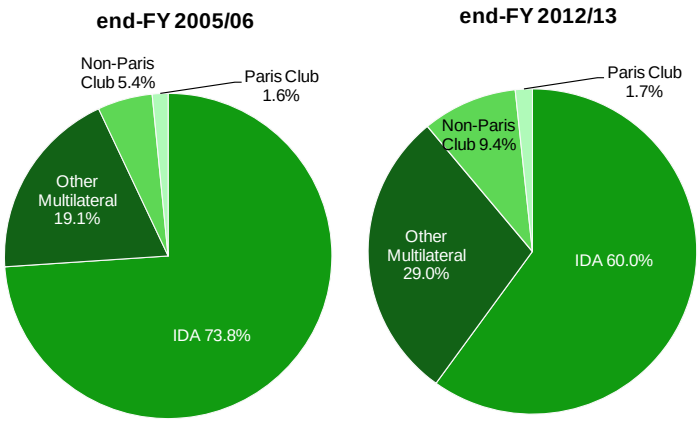
Table 1. Post-MDRI Debt Accumulation: Uganda Versus Selected Post-MDRI Countries[1]

| | External Debt | | | Domestic Debt | | | Public Debt | | |
|---|---|---|---|---|---|---|---|---|---|
| | 2007 | 2012 | Change | 2007 | 2012 | Change | 2007 | 2012 | Change |
| | | | | (In percent of GDP) | | | | | |
| Zambia | 10.3 | 16.3 | 6.0 | 16.4 | 16.1 | -0.3 | 26.7 | 32.4 | 5.6 |
| Ethiopia | 11.3 | 17.9 | 6.7 | 25.9 | 3.3 | -22.6 | 37.2 | 21.2 | -16.0 |
| **Uganda** | **12.3** | **17.1** | **4.8** | **9.0** | **13.1** | **4.1** | **21.3** | **30.3** | **8.9** |
| Benin | 12.7 | 17.0 | 4.3 | 8.5 | 12.1 | 3.6 | 21.2 | 29.1 | 7.9 |
| Ghana | 14.5 | 21.9 | 7.4 | 16.6 | 28.3 | 11.8 | 31.0 | 50.2 | 19.1 |
| Rwanda | 15.7 | 14.5 | -1.3 | 11.9 | 9.6 | -2.3 | 27.6 | 24.1 | -3.5 |
| Niger | 16.0 | 17.7 | 1.7 | 9.3 | 11.0 | 1.7 | 25.4 | 28.8 | 3.4 |
| Honduras | 16.4 | 19.7 | 3.3 | 8.2 | 14.7 | 6.5 | 24.6 | 34.4 | 9.8 |
| Mali | 18.1 | 26.3 | 8.2 | 3.0 | 3.5 | 0.4 | 21.1 | 29.7 | 8.6 |
| Senegal | 19.0 | 31.3 | 12.3 | 4.4 | 10.3 | 5.9 | 23.5 | 41.7 | 18.2 |
| Tanzania | 19.6 | 29.7 | 10.1 | 8.8 | 11.1 | 2.2 | 28.4 | 40.8 | 12.3 |
| Burkina Faso | 23.3 | 23.8 | 0.4 | 2.1 | 3.5 | 1.5 | 25.4 | 27.3 | 1.9 |
| Madagascar | 26.0 | 25.8 | -0.1 | 7.5 | 12.3 | 4.8 | 33.5 | 38.1 | 4.7 |
| Mozambique | 40.8 | 34.8 | -6.1 | 1.1 | 7.5 | 6.4 | 41.9 | 42.2 | 0.3 |
| Guyana | 41.3 | 48.0 | 6.7 | 18.6 | 17.2 | -1.3 | 59.9 | 65.3 | 5.4 |
| **All-median** | **16.4** | **21.9** | **4.8** | **8.8** | **11.1** | **2.2** | **26.7** | **32.4** | **5.6** |
| **Low post-MDRI debt [2]** | **15.7** | **17.9** | **6.0** | **9.0** | **11.1** | **2.2** | **25.4** | **30.3** | **8.6** |
| **High post-MDRI debt [2]** | **33.4** | **30.3** | **0.2** | **4.8** | **9.9** | **3.1** | **37.7** | **40.2** | **3.3** |
| *Uganda percentiles* | | | | | | | | | |
| *All post-MDRI countries* | *14* | *21* | *50* | *57* | *71* | *64* | *14* | *43* | *71* |
| *Low post-MDRI debt [2]* | *20* | *30* | *40* | *50* | *70* | *70* | *20* | *50* | *60* |

[1] For comparability to Uganda's Debt ratios countries that received the MDRI relief in 2006 are reported.
[2] Median. Low post-MDRI debt refers to countries with external debt lower than 20 percent of GDP.

**4. Concessional multilateral debt continues to constitute most of Uganda's public and publicly guaranteed (PPG) external debt.** The share of multilateral creditors—primarily the World Bank and the African Development Bank—remains close to 90 percent, only marginally lower than their pre-MDRI share. Owing almost exclusively to higher borrowing from China, non-Paris Club creditors account for close to 10 percent of PPG external debt. During the previous PSI Uganda accumulated only $455 million in NCB, substantially lower than the ceiling of $1 billion.

Composition of PPG External Debt




**5. Despite a recent increase, private sector debt remains relatively low.** Preliminary data shows that after remaining stable at about 7 percent of GDP in FY2000–2008, private sector external debt has increased to about to 16¼ percent of GDP in FY2012. The upsurge was driven by intercompany lending, which now represents more than half of private external debt.

**6. The authorities have taken steps to improve debt management.** The government is undertaking regular assessments of debt maturities and associated risks, notably exchange rate and rollover risks. In line with its revised debt strategy, in 2013 the government started restructuring debt management processes and the institutional framework, and is planning to set up a debt management unit within the Ministry of Finance. The new medium-term debt strategy accounts for a planned scaling up of NCB to finance key infrastructure projects.

## Underlying Assumptions

**7. This DSA is consistent with the macroeconomic framework outlined in the main report.** The baseline scenario assumes implementation of the economic and structural policies envisaged in the PSI. Expected gains in revenue mobilization, further deepening of domestic debt markets, and greater availability of financing at non-concessional but still reasonable terms are expected to create the fiscal space for significant scaling up of infrastructure investment—mainly on energy generation and road construction. Taking into account projects starting later this fiscal year (construction of the Karuma and Isimba dams), and later on oil-related infrastructure, NCB will need to increase from $1.5 billion to $2.2 billion in the program period. Projections for key macroeconomic variables are explained below. Box 1 presents changes in assumptions in comparison to the previous DSA.

## Macroeconomic Assumptions

- **Average medium-term growth** is projected to increase to about 7 percent of GDP mainly supported by infrastructure investment. Once the large energy and road projects are completed and ease the acutely binding infrastructure gap, potential output is likely to be

higher, particularly if supported by an environment conducive to private sector development. On this basis, average long-term growth is projected at 7.2 percent.

- **Over the medium-term the average current account deficit** is projected to reach 12½ percent of GDP, reflecting high import demand associated with the construction of large infrastructure projects. Once these are completed, import demand would come down while export receipts would increase on account of potential energy surpluses and better road and railway infrastructure. As a result, under the baseline scenario, which does not include oil exports, the current account deficit is projected, on average, at 7½ percent of GDP in the long term.

- **The fiscal deficit** is projected to widen from about 4 percent of GDP in FY2012/13 to 5¼ percent of GDP in the medium term, peaking at 7½ percent of GDP in FY2013/14. This increase reveals the impact of the construction of large projects. The long-term average is expected to come down to about 2¾ percent of GDP.

## Other Assumptions

- **An increase in the NCB** ceiling from $1.5 billion to $2.213 billion in is incorporated. While debt will continue to be primarily highly concessional, additional concessional resources are not available to finance the large projects needed for addressing the infrastructure deficit. Uganda ranks 133th out of 146 countries in indicators on quality of infrastructure, in particular roads, railroads, and availability of electricity. It also lags behind most of its EAC partners. The World Bank supports the authorities' decision to develop critical infrastructure projects.

| | | | *Quality of infrastructure* | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Infrastructure | | Roads | | Railroads | | Electricity and telephony | |
| | Score | Ranking | Score | Ranking | Score | Ranking | Score | Ranking |
| **Uganda** | **2.27** | **133** | **2.91** | **110** | **1.44** | **111** | **1.80** | **133** |
| Kenya | 3.09 | 103 | 3.93 | 72 | 2.50 | 72 | 2.73 | 111 |
| Tanzania | 2.27 | 132 | 3.22 | 94 | 2.26 | 82 | 1.77 | 134 |
| Rwanda | 3.22 | 96 | 4.96 | 40 | … | … | 2.71 | 112 |
| Burundi | 1.87 | 141 | 2.70 | 121 | … | … | 1.44 | 139 |
| **Median (all countries)** | **3.91** | **73** | **3.93** | **73** | **2.66** | **63** | **4.24** | **73** |

Source: World Economic Forum.

- **Improvements in implementation capacity** will be critical for the planned scaling up of infrastructure spending. Uganda's weak track record of implementation of public investment raises concerns. However, recent improvements in implementation of road projects, as well as prospects for tapping into technical and project management expertise of external contractors and, when needed, high-skilled workers are likely to alleviate technical capacity constraints.

- **Another key consideration is preserving macroeconomic stability.** The expected high import content would significantly limit inflationary and exchange rate appreciation pressures. Staff estimates that the growth stimulus associated with the projects would be compatible with potential output and not lead to overheating concerns.

**Box 1.Uganda: Assumptions in Comparison with the Previous DSA**

Compared to the previous joint IMF/IDA DSA, conducted in the context of the sixth review under the PSI, key changes are as follows:

- **Non-concessional borrowing (NCB).** The NCB ceiling is revised upward from $1.5 billion to $2.2 billion to reflect the change in the financing modalities of the Karuma HPP, the construction of industrial substations for transmission, and smaller-scale oil-related infrastructure projects.
- **Discount rate.** An increase from 3 percent to 5 percent, in line with the new guidelines, leads to an increase in the grant element of both the new borrowing and the debt service associated with the existing debt, thereby reducing the net present value of debt.
- **Financing terms for the NCB.** Based on prospective financing terms of the Eximbank China loans under negotiation, staff estimates a grant element of about 11½ percent in the financing package of the Karuma, Isimba, and industrial substation projects. Overall, the DSA incorporates a higher grant element for new borrowing compared to the 6th review, taking into account the increase in the discount rate and the impact on interest rates of the expected tightening of global liquidity conditions over the medium-term.
- **Growth.** The positive impact of the projects on the infrastructure gap, together with updated developments in the domestic and global economy, led staff to raise long-run growth projections from 7 percent to 7¼ percent.
- **Contingent liabilities from PPPs.** Expected contingent liabilities associated with two road projects to be developed under PPP arrangements, amounting to about 1½ percent of GDP, are included in the baseline projections.
- **Domestic borrowing.** With the recent extension of effective maturity to close to 2.7 years, the annual rollover of domestic debt stock has been reduced to about 35 percent from the previous 50 percent, reducing the rollover risk of public debt service.

## B. External Debt Sustainability Analysis

**8. PPG external debt is assessed to be sustainable over the projection period.** All debt burden indicators are projected to remain below Uganda's country-specific debt burden thresholds under the baseline scenario and the standardized stress tests (Figure 1a, Table 1a, and Table 1b).

**9. Uganda's external debt is projected to remain below the relevant debt burden thresholds even if long-term growth rates weaken sharply.** To examine the sensitivity of DSA results to changes in long-term GDP growth, staff examined a shock scenario that illustrates at which long-term growth rate the baseline projections would exceed relevant external debt thresholds. The results show that an implausibly large decline in long-term growth from 7¼ percent to 2½ percent would be required to breach the debt service-to-revenue threshold by the end of the projection period. However, these results should be interpreted cautiously as this partial-equilibrium analysis does not incorporate feedback effects of a sharp and persistent deceleration in output on key macroeconomic aggregates including the fiscal deficit.

**10. An alternative scenario that considers prospective construction of a regional railway shows that debt would still remain safely within the relevant thresholds.** This scenario assesses implications of additional NCB of about $1.6 billion beyond the PSI period to finance the Ugandan share of the Mombasa-Kampala-Kigali regional railway project. Debt burden indicators, while significantly higher, would still be sustainable.

**11. An illustrative scenario that includes the onset of oil production beginning in 2018 shows that the debt outlook would improve significantly (Box 2).** This scenario simply illustrates the impact of oil revenues on the DSA, keeping expenditures unchanged with respect to the baseline. The borrowing needs for oil sector development would lead to an initial marginal increase in debt. Not surprisingly, when oil revenues come on stream, the DSA is projected to improve significantly with an elimination of external debt by 2025. This illustrates the space available to Uganda to use a significant portion of its prospective oil revenue to finance infrastructure investment.

### Box 2. Assumptions for the Oil Illustrative Scenario

**Cost and financing.** Construction of the envisaged pipeline and small refinery is planned to start in FY2015/16. Total capital expenditure is estimated at $15-$20 billion, and would be financed primarily through FDI. The government will need to increase external borrowing only marginally to meet the expected equity commitments in both projects.

**Growth impact.** Based on the investment and production profile, real GDP growth in 2016–2023 is expected to be 2–4 percent higher than under the baseline scenario. Growth would pick up during the investment phase and the onset of full production. Oil production is expected to account for close to 15 percent of Uganda's GDP during the peak extraction period. Oil reserves are anticipated to last for about 30 years.

**Revenue impact.** Oil revenues would amount to more than 50 percent of total government revenues. For the exercise, oil export prices are conservatively assumed to be constant in real terms at 80 dollars a barrel. Domestically refined oil is priced at import-parity minus a 20 percent discount.

## C. Public Debt Sustainability Update

**12. The evolution of total public debt (external and domestic debt) is sustainable over the projection period under the baseline scenario and when subject to stress tests.** The public debt-to-GDP ratio is projected to peak at about 40 percent of GDP in 2018, well below the benchmark level of 74 percent associated with heightened public debt vulnerabilities for strong performers.[2] However, the relatively short average maturity of domestic debt

[2] Uganda is ranked as a "strong performer" under the Country Policy and Institutional Assessment (CPIA) framework of the World Bank. Accordingly, debt burden thresholds applicable for Uganda are PV of debt to GDP ratio of 50 percent, PV value of debt–to-exports ratio of 200 percent, PV of debt-to-revenue ratio of 300 percent, debt-service-to-exports ratio of 25 percent, and debt-service-to-revenue ratio of 22 percent. The indicative benchmark for public-debt-to-GDP ratio is 74 percent.

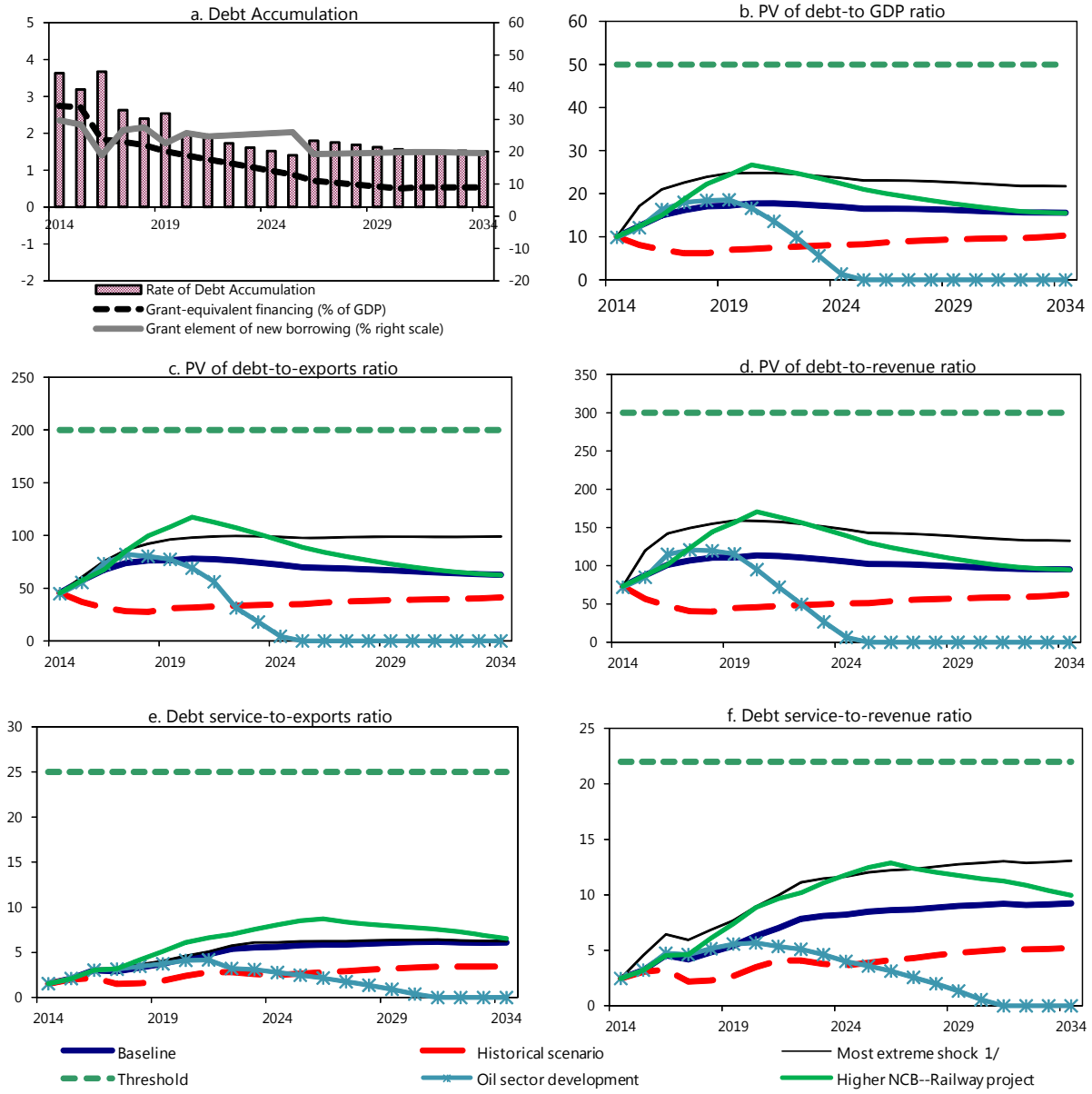
(less than three years) combined with a low revenue base leads to a debt service-to-revenue ratio of about 35 percent, among the highest in LICs. This significantly increases the rollover and interest rate risks, and needs to be addressed in the medium term by a combination of stronger revenue mobilization and deeper financial markets to extend average maturities.

**13. Stress tests indicate that the path of public debt would become unsustainable in the absence of fiscal consolidation (Table 2b).** In an illustrative scenario that assumes an unchanged primary deficit over the projection period at 5.9 percent of GDP, the NPV of public debt to GDP would grow rapidly and reach 77 percent by 2034. These results highlight the importance of reducing fiscal deficits after the temporary increase during the scaling up of public investment.

## D. Conclusion

**14. Despite the envisaged scaling-up of external borrowing, Uganda continues to face a low risk of debt distress.** The government's cautious approach in accumulating new external debt during the post-MDRI period has provided the economy with significant borrowing space to scale up public investment. However, the authorities are encouraged to adhere to the planned pace of implementation of these projects and ensure appropriate cost recovery to avoid delays and inefficiencies that could add costs and affect economic stability. Once the construction of the projects is completed, the temporary increase in fiscal deficits should be halted to bring public debt back to a sustainable path. The planned increase in tax revenues and maintenance of a stable economic environment should help reduce the existing rollover and interest risk of domestic debt.

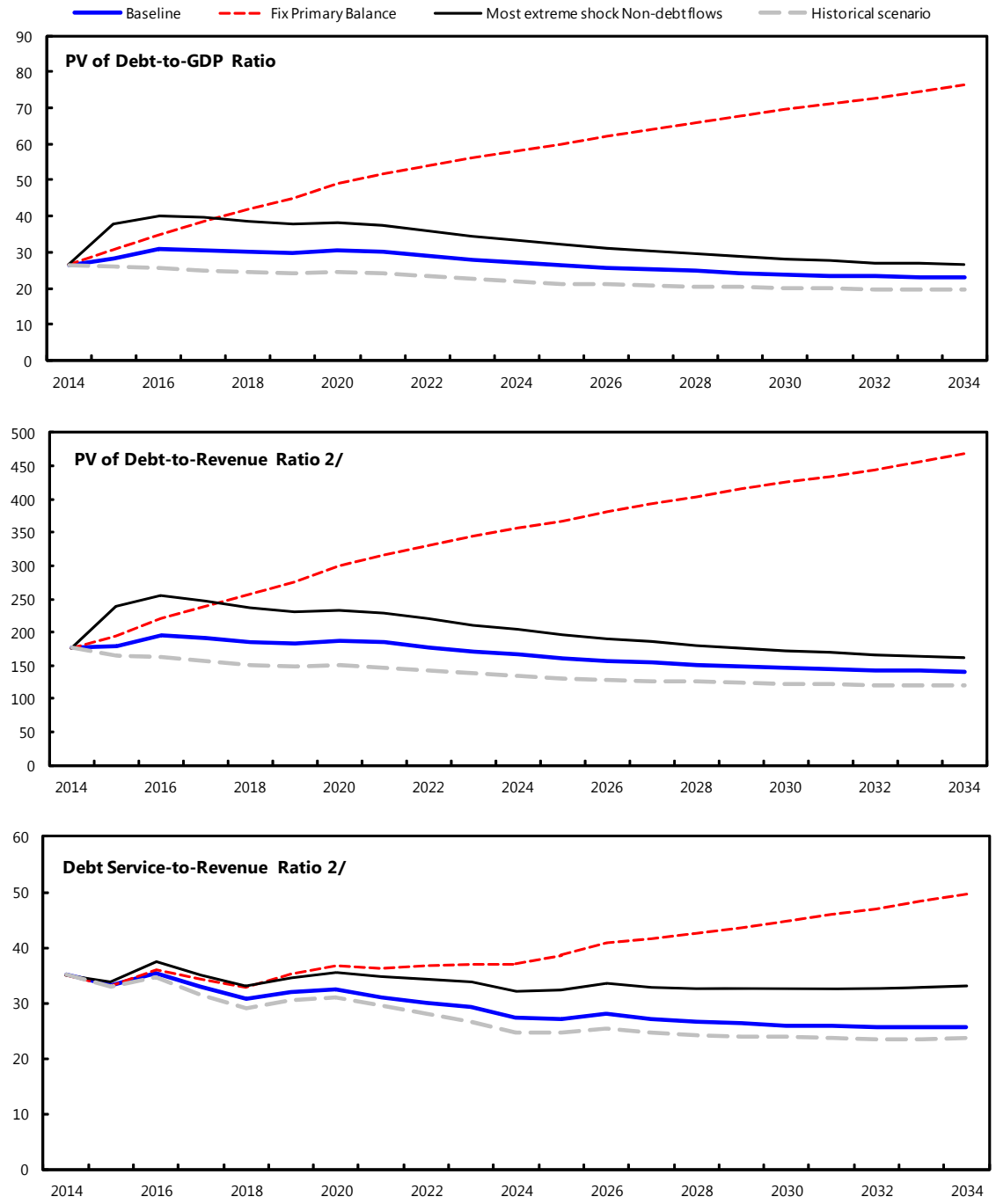
**Figure 1.a. Uganda: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2014–2034[1]**




Sources: Uganda authorities; and IMF staff estimates and projections.

1/ The most extreme stess test is the test that yields the highest ratio in 2024. In figure b. it corresponds to a One-time depreciation shock; in c. to a Terms shock; in d. to a One-time depreciation shock; in e. to a Terms shock and in figure f. to a One-time depreciation shock.

**Figure 1.b. Uganda: Indicators of Public Debt Under Alternatives Scenarios, 2014–2034[1]**




Sources: Uganda authorities; and IMF staff estimates and projections.
1/ The most extreme stress tests is the test that yields the highest ratio in 2014
2/ Revenues are defined inclusive of grants.

## Table 1a. Uganda: Public Sector Debt Sustainability Framework, Baseline Scenario, 2011–2034

| | Actual | | | | | Estimate | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | Average 5/ | Standard Deviation 5/ | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2014-19 Average | 2024 | 2034 | 2020-34 Average |
| **Public sector debt 1/** | 31.9 | 29.0 | 32.8 | | | 36.4 | 38.5 | 41.0 | 40.6 | 39.8 | 38.9 | | 34.7 | 28.5 | |
| *Of which* :foreign-currency denominated | 18.2 | 15.6 | 17.5 | | | 19.9 | 22.8 | 25.2 | 26.2 | 26.9 | 26.4 | | 24.5 | 21.0 | |
| Change in public sector debt | 7.3 | -2.9 | 3.8 | | | 3.6 | 2.1 | 2.5 | -0.4 | -0.8 | -0.9 | | -1.2 | -0.2 | |
| Identified debt-creating flows | 3.8 | -4.8 | 2.0 | | | 5.0 | 1.8 | 2.4 | 0.0 | -0.6 | -0.9 | | -1.2 | -0.2 | |
| Primary deficit | 3.3 | 1.8 | 2.5 | 1.1 | 1.8 | 5.9 | 3.3 | 4.0 | 1.9 | 1.3 | 1.5 | 3.0 | 1.3 | 1.2 | 1.2 |
| Revenue and grants | 18.4 | 15.5 | 14.8 | | | 15.1 | 15.8 | 15.8 | 16.0 | 16.3 | 16.4 | | 16.4 | 16.4 | |
| *of which: grants* | 2.3 | 2.3 | 1.7 | | | 1.4 | 1.5 | 1.0 | 0.9 | 0.8 | 0.8 | | 0.3 | 0.0 | |
| Primary (noninterest) expenditure | 21.7 | 17.3 | 17.3 | | | 20.9 | 19.1 | 19.8 | 17.9 | 17.6 | 17.9 | | 17.6 | 17.6 | |
| Automatic debt dynamics | 0.5 | -6.7 | -0.5 | | | -2.3 | -1.6 | -1.7 | -1.9 | -1.9 | -2.4 | | -2.5 | -1.4 | |
| Contribution from interest rate/growth differential | -1.0 | -2.7 | -0.9 | | | -1.8 | -2.0 | -1.8 | -1.9 | -1.9 | -1.4 | | -2.1 | -1.2 | |
| *of which: contribution from average real interest rate* | 0.5 | -1.6 | 0.7 | | | 0.2 | 0.3 | 0.8 | 0.8 | 0.8 | 1.3 | | 0.3 | 0.7 | |
| *of which: contribution from real GDP growth* | -1.5 | -1.0 | -1.6 | | | -1.9 | -2.2 | -2.5 | -2.7 | -2.7 | -2.7 | | -2.4 | -1.9 | |
| Contribution from real exchange rate depreciation | 1.4 | -4.0 | 0.3 | | | -0.6 | 0.4 | 0.1 | 0.1 | 0.0 | -1.0 | | ... | ... | |
| Other identified debt-creating flows | 0.0 | 0.0 | 0.0 | | | 1.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 1.5 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | 3.5 | 2.0 | 1.8 | | | -1.4 | 0.3 | 0.1 | -0.4 | -0.2 | 0.0 | | 0.0 | 0.0 | |
| **Other Sustainability Indicators** | | | | | | | | | | | | | | | |
| **PV of public sector debt** | ... | ... | 23.0 | | | 26.6 | 28.2 | 30.8 | 30.6 | 30.0 | 29.8 | | 27.2 | 23.1 | |
| *Of which* : foreign-currency denominated | ... | ... | 7.6 | | | 10.0 | 12.5 | 15.0 | 16.2 | 17.0 | 17.2 | | 16.9 | 15.6 | |
| *Of which* : external | ... | ... | 7.6 | | | 10.0 | 12.5 | 15.0 | 16.2 | 17.0 | 17.2 | | 16.9 | 15.6 | |
| PV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need 2/ | 9.9 | 6.5 | 12.5 | | | 16.5 | 14.3 | 15.1 | 12.6 | 11.2 | 11.2 | | 9.2 | 7.6 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | ... | ... | 155.0 | | | 176.0 | 178.4 | 195.3 | 190.7 | 184.4 | 182.1 | | 166.0 | 141.1 | |
| PV of public sector debt-to-revenue ratio (in percent) | ... | ... | 174.9 | | | 193.8 | 197.6 | 208.6 | 202.3 | 194.4 | 191.7 | | 169.3 | 141.1 | |
| *Of which* : external 3/ | ... | ... | 58.1 | | | 73.2 | 87.6 | 101.4 | 106.9 | 110.6 | 110.9 | | 105.7 | 95.3 | |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 33.7 | 28.4 | 35.1 | | | 35.1 | 33.1 | 35.3 | 33.0 | 30.7 | 31.9 | | 27.3 | 25.5 | |
| Debt service-to-revenue ratio (in percent) 4/ | 38.4 | 33.3 | 39.6 | | | 38.6 | 36.6 | 37.7 | 35.0 | 32.4 | 33.6 | | 27.8 | 25.5 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | -4.0 | 4.7 | -1.3 | | | 2.2 | 1.2 | 1.6 | 2.2 | 2.1 | 2.4 | | 2.5 | 1.4 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 6.6 | 3.4 | 5.8 | 7.0 | 2.0 | 6.2 | 6.5 | 7.1 | 7.2 | 7.2 | 7.2 | 6.9 | 7.2 | 7.2 | 7.2 |
| Average nominal interest rate on forex debt (in percent) | 1.0 | 1.1 | 1.1 | 1.1 | 0.5 | 1.1 | 1.5 | 2.1 | 2.1 | 2.2 | 2.4 | 1.9 | 2.6 | 3.1 | 2.7 |
| Average real interest rate on domestic debt (in percent) | 5.5 | -11.7 | 6.8 | 4.4 | 8.2 | 2.2 | 2.2 | 5.1 | 5.4 | 5.4 | 9.6 | 5.0 | 1.5 | 6.5 | 4.8 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | 10.1 | -23.0 | 2.1 | -2.9 | 12.1 | -3.6 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 5.0 | 24.1 | 4.7 | 8.7 | 6.4 | 7.8 | 6.9 | 4.7 | 4.8 | 5.0 | 3.4 | 5.4 | 6.8 | 5.8 | 6.4 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | 0.3 | -0.2 | 0.1 | 0.1 | 0.1 | 0.3 | 0.0 | 0.1 | 0.0 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | ... | ... | 29.7 | 28.4 | 18.9 | 26.7 | 27.5 | 22.7 | 25.7 | 25.8 | 19.6 | ... |

Sources: Ugandan authorities and IMF staff estimates and projections.
1/ Public sector includes general government only and gross debt is used for all presentations.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

## Table 1b. Uganda: External Debt Sustainability Framework, baseline Scenario, 2011–2013[1]

| | Actual | | | Historical Average 6/ | Standard Deviation 6/ | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2011 | 2012 | 2013 | | | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2014-2019 Average | 2024 | 2034 | 2020-2034 Average |
| **External debt (nominal) 1/** | **32.9** | **31.9** | **36.4** | | | **39.9** | **43.1** | **46.5** | **48.0** | **48.5** | **47.4** | | **43.5** | **40.0** | |
| o/w public and publicly guaranteed (PPG) | 18.2 | 15.6 | 17.5 | | | 19.9 | 22.8 | 25.2 | 26.2 | 26.9 | 26.4 | 9.4 | 24.5 | 21.0 | |
| Change in external debt | 6.2 | -0.9 | 4.4 | | | 3.5 | 3.2 | 3.4 | 1.5 | 0.5 | -1.1 | | -0.6 | -0.1 | |
| Identified net debt-creating flows | 8.1 | 1.5 | 2.9 | | | 7.1 | 4.4 | 3.9 | 2.5 | 1.1 | 0.2 | | 0.5 | 0.5 | |
| **Non-interest current account deficit** | **11.5** | **12.9** | **9.6** | **6.8** | **4.2** | **13.1** | **11.9** | **12.4** | **11.7** | **10.1** | **7.2** | | **7.0** | **6.2** | 6.9 |
| Deficit in balance of goods and services | 18.3 | 15.5 | 11.4 | | | 14.6 | 13.7 | 14.2 | 13.2 | 11.7 | 11.5 | | 10.5 | 9.2 | |
| Exports | 22.3 | 23.5 | 23.8 | | | 22.0 | 21.8 | 22.2 | 21.9 | 22.3 | 22.5 | | 23.5 | 24.8 | |
| Imports | 40.6 | 39.0 | 35.2 | | | 36.6 | 35.5 | 36.4 | 35.0 | 34.0 | 34.0 | | 34.0 | 34.0 | |
| Net current transfers (negative = inflow) | -8.5 | -6.3 | -6.0 | -7.8 | 1.5 | -5.5 | -5.4 | -5.4 | -5.3 | -5.2 | -5.1 | | -4.3 | -3.7 | -4.1 |
| o/w official | -4.0 | -2.0 | -0.4 | | | -0.3 | -0.3 | -0.3 | -0.2 | -0.2 | -0.2 | | -0.2 | 0.0 | |
| Other current account flows (negative = net inflow) | 1.7 | 3.7 | 4.2 | | | 3.9 | 3.6 | 3.6 | 3.8 | 3.7 | 0.8 | | 0.8 | 0.8 | |
| **Net FDI (negative = inflow)** | **-4.3** | **-7.1** | **-4.3** | **-4.8** | **1.1** | **-4.4** | **-5.6** | **-6.3** | **-6.9** | **-6.6** | **-4.6** | | **-4.4** | **-3.9** | -4.2 |
| **Endogenous debt dynamics 2/** | **0.9** | **-4.2** | **-2.3** | | | **-1.6** | **-1.9** | **-2.3** | **-2.4** | **-2.5** | **-2.4** | | **-2.1** | **-1.9** | |
| Contribution from nominal interest rate | 0.3 | 0.5 | 0.4 | | | 0.4 | 0.5 | 0.5 | 0.7 | 0.7 | 0.7 | | 0.7 | 0.7 | |
| Contribution from real GDP growth | -1.8 | -1.0 | -1.7 | | | -2.0 | -2.4 | -2.9 | -3.0 | -3.2 | -3.1 | | -2.9 | -2.6 | |
| Contribution from price and exchange rate changes | 2.4 | -3.7 | -1.0 | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| **Residual (3-4) 3/** | **-1.9** | **-2.4** | **1.5** | | | **-3.6** | **-1.2** | **-0.5** | **-1.0** | **-0.6** | **-1.3** | | **-1.1** | **-0.6** | |
| o/w exceptional financing | 0.0 | 0.0 | 0.1 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| PV of external debt 4/ | ... | ... | 26.6 | | | 30.0 | 32.8 | 36.3 | 38.0 | 38.6 | 38.2 | 11.7 | 35.9 | 34.6 | |
| In percent of exports | ... | ... | 111.6 | | | 136.3 | 150.4 | 163.4 | 173.5 | 173.1 | 169.8 | | 153.0 | 139.3 | |
| **PV of PPG external debt** | **...** | **...** | **7.6** | | | **10.0** | **12.5** | **15.0** | **16.2** | **17.0** | **17.2** | | **16.9** | **15.6** | |
| **In percent of exports** | **...** | **...** | **32.1** | | | **45.6** | **57.3** | **67.5** | **73.9** | **76.3** | **76.5** | | **72.2** | **62.8** | |
| **In percent of government revenues** | **...** | **...** | **58.1** | | | **73.2** | **87.6** | **101.4** | **106.9** | **110.6** | **110.9** | | **105.7** | **95.3** | |
| **Debt service-to-exports ratio (in percent)** | **4.4** | **6.2** | **5.0** | | | **4.5** | **5.1** | **6.7** | **6.8** | **6.9** | **7.0** | | **8.5** | **9.0** | |
| **PPG debt service-to-exports ratio (in percent)** | **2.4** | **2.1** | **2.0** | | | **1.5** | **2.1** | **3.0** | **2.9** | **3.3** | **3.7** | | **5.6** | **6.1** | |
| **PPG debt service-to-revenue ratio (in percent)** | **3.3** | **3.7** | **3.6** | | | **2.5** | **3.2** | **4.5** | **4.2** | **4.8** | **5.4** | | **8.2** | **9.2** | |
| Total gross financing need (Billions of U.S. dollars) | 1.4 | 1.4 | 1.4 | | | 2.4 | 2.0 | 2.2 | 2.0 | 1.7 | 1.5 | | 2.9 | 8.1 | |
| Non-interest current account deficit that stabilizes debt ratio | 5.3 | 13.8 | 5.1 | | | 9.5 | 8.7 | 9.1 | 10.2 | 9.7 | 8.3 | | 7.6 | 6.3 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 6.6 | 3.4 | 5.8 | 7.0 | 2.0 | 6.2 | 6.5 | 7.1 | 7.2 | 7.2 | 7.2 | 6.9 | 7.2 | 7.2 | 7.2 |
| GDP deflator in US dollar terms (change in percent) | -8.3 | 12.8 | 3.3 | 5.5 | 8.3 | 7.4 | 1.5 | -0.7 | 2.1 | 1.5 | 3.6 | 2.6 | 3.6 | 2.7 | 3.5 |
| Effective interest rate (percent) 5/ | 1.1 | 1.7 | 1.3 | 1.9 | 1.1 | 1.1 | 1.3 | 1.3 | 1.6 | 1.6 | 1.7 | 1.4 | 1.8 | 2.0 | 1.9 |
| Growth of exports of G&S (US dollar terms, in percent) | 6.3 | 22.7 | 10.7 | 21.5 | 10.3 | 5.7 | 7.0 | 8.2 | 7.9 | 11.0 | 12.0 | 8.6 | 12.0 | 11.0 | 11.6 |
| Growth of imports of G&S (US dollar terms, in percent) | 18.8 | 12.0 | -1.5 | 17.1 | 12.8 | 19.0 | 4.8 | 9.0 | 5.3 | 5.5 | 11.1 | 9.1 | 11.1 | 10.0 | 10.9 |
| Grant element of new public sector borrowing (in percent) | ... | ... | ... | ... | ... | 29.7 | 28.4 | 18.9 | 26.7 | 27.5 | 22.7 | 25.7 | 25.8 | 19.6 | 22.0 |
| Government revenues (excluding grants, in percent of GDP) | 16.2 | 13.2 | 13.2 | | | 13.7 | 14.3 | 14.8 | 15.1 | 15.4 | 15.5 | | 16.0 | 16.4 | 16.1 |
| Aid flows (in Billions of US dollars) 7/ | 0.7 | 0.8 | 1.0 | | | 0.9 | 0.9 | 0.6 | 0.7 | 0.7 | 0.7 | | 0.9 | 1.4 | |
| o/w Grants | 0.4 | 0.4 | 0.4 | | | 0.3 | 0.4 | 0.3 | 0.3 | 0.3 | 0.3 | | 0.2 | 0.0 | |
| o/w Concessional loans | 0.3 | 0.3 | 0.6 | | | 0.5 | 0.5 | 0.3 | 0.4 | 0.5 | 0.4 | | 0.7 | 1.4 | |
| Grant-equivalent financing (in percent of GDP) 8/ | ... | ... | ... | | | 2.7 | 2.7 | 1.8 | 1.8 | 1.7 | 1.5 | | 1.0 | 0.5 | 0.8 |
| Grant-equivalent financing (in percent of external financing) 8/ | ... | ... | ... | | | 46.1 | 47.8 | 34.0 | 43.0 | 43.1 | 38.9 | | 33.9 | 19.6 | 27.0 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Nominal GDP (Billions of US dollars) | 16.8 | 19.6 | 21.4 | | | 24.5 | 26.5 | 28.1 | 30.8 | 33.5 | 37.2 | | 62.8 | 176.1 | |
| Nominal dollar GDP growth | -2.2 | 16.6 | 9.3 | | | 14.1 | 8.1 | 6.3 | 9.4 | 8.8 | 11.1 | 9.6 | 11.1 | 10.0 | 10.9 |
| PV of PPG external debt (in Billions of US dollars) | | | 1.6 | | | 2.4 | 3.2 | 4.2 | 4.9 | 5.6 | 6.5 | | 10.5 | 27.0 | |
| (PVt-PVt-1)/GDPt-1 (in percent) | | | | | | 3.6 | 3.2 | 3.7 | 2.6 | 2.4 | 2.5 | 3.0 | 1.5 | 1.5 | 1.6 |
| Gross workers' remittances (Billions of US dollars) | 0.8 | 0.8 | 1.2 | | | 1.3 | 1.3 | 1.4 | 1.5 | 1.7 | 1.8 | | 3.0 | 7.5 | |
| PV of PPG external debt (in percent of GDP + remittances) | ... | ... | 7.2 | | | 9.5 | 11.9 | 14.3 | 15.4 | 16.2 | 16.4 | | 16.2 | 14.9 | |
| PV of PPG external debt (in percent of exports + remittances) | ... | ... | 26.0 | | | 36.9 | 46.5 | 54.9 | 60.1 | 62.4 | 62.7 | | 60.1 | 53.6 | |
| Debt service of PPG external debt (in percent of exports + remittances) | ... | ... | 1.6 | | | 1.2 | 1.7 | 2.5 | 2.4 | 2.7 | 3.1 | | 4.7 | 5.2 | |

Sources: Ugandan authorities; and IMF staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

## Table 2a. Uganda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034

(in Percent)

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | **2024** | 2034 |
| **PV of debt-to GDP ratio** | | | | | | | | |
| **Baseline** | 10 | 13 | 15 | 16 | 17 | 17 | **17** | 16 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 10 | 8 | 7 | 6 | 6 | 7 | **8** | 10 |
| A2. New public sector loans on less favorable terms in 2014-2034 2 | 10 | 13 | 17 | 19 | 21 | 22 | **23** | 25 |
| A3. Oil sector development starting from FY2013/14 | 10 | 12 | 16 | 18 | 18 | 19 | **1** | -12 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 10 | 12 | 15 | 16 | 17 | 18 | **17** | 16 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 10 | 11 | 13 | 14 | 15 | 16 | **16** | 15 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 10 | 13 | 16 | 17 | 18 | 19 | **18** | 16 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 10 | 13 | 17 | 18 | 19 | 19 | **18** | 16 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 10 | 10 | 10 | 11 | 12 | 13 | **14** | 15 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 10 | 17 | 21 | 23 | 24 | 25 | **24** | 22 |
| **PV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 46 | 57 | 67 | 74 | 76 | 77 | **72** | 63 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 46 | 37 | 32 | 28 | 28 | 31 | **35** | 41 |
| A2. New public sector loans on less favorable terms in 2014-2034 2 | 46 | 61 | 76 | 86 | 92 | 96 | **99** | 99 |
| A3. Oil sector development starting from FY2013/14 | 45 | 55 | 73 | 82 | 80 | 77 | **4** | -43 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 46 | 55 | 67 | 73 | 75 | 77 | **71** | 62 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 46 | 50 | 54 | 60 | 63 | 65 | **62** | 57 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 46 | 55 | 67 | 73 | 75 | 77 | **71** | 62 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 46 | 59 | 76 | 82 | 84 | 86 | **76** | 63 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 46 | 43 | 39 | 44 | 48 | 51 | **51** | 51 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 46 | 55 | 67 | 73 | 75 | 77 | **71** | 62 |
| **PV of debt-to-revenue ratio** | | | | | | | | |
| **Baseline** | 73 | 88 | 101 | 107 | 111 | 111 | **106** | 95 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 73 | 57 | 48 | 41 | 40 | 45 | **51** | 63 |
| A2. New public sector loans on less favorable terms in 2014-2034 2 | 73 | 93 | 114 | 125 | 133 | 139 | **144** | 150 |
| A3. Oil sector development starting from FY2013/14 | 72 | 85 | 115 | 121 | 120 | 116 | **6** | -68 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 73 | 86 | 104 | 109 | 113 | 116 | **107** | 97 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 73 | 80 | 87 | 93 | 97 | 101 | **97** | 92 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 73 | 88 | 107 | 112 | 117 | 120 | **111** | 100 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 73 | 90 | 115 | 119 | 122 | 124 | **111** | 95 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 73 | 72 | 68 | 75 | 81 | 86 | **87** | 90 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 73 | 120 | 142 | 149 | 155 | 159 | **148** | 133 |

## Table 2a. Uganda: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–2034 (continued)

(in Percent)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **Debt service-to-exports ratio** | | | | | | | | |
| **Baseline** | 2 | 2 | 3 | 3 | 3 | 4 | **6** | 6 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 2 | 2 | 2 | 2 | 2 | 2 | **2** | 3 |
| A2. New public sector loans on less favorable terms in 2014-2034 2 | 2 | 2 | 2 | 3 | 3 | 4 | **6** | 8 |
| A3. Oil sector development starting from FY2013/14 | 2 | 2 | 3 | 3 | 3 | 4 | **3** | -1 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 2 | 2 | 3 | 3 | 3 | 4 | **6** | 6 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 2 | 2 | 3 | 2 | 3 | 3 | **5** | 6 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 2 | 2 | 3 | 3 | 3 | 4 | **6** | 6 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 2 | 2 | 3 | 3 | 4 | 4 | **6** | 6 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 2 | 2 | 2 | 2 | 2 | 3 | **4** | 5 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 2 | 2 | 3 | 3 | 3 | 4 | **6** | 6 |
| **Debt service-to-revenue ratio** | | | | | | | | |
| **Baseline** | 2 | 3 | 5 | 4 | 5 | 5 | **8** | 9 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2014-2034 1/ | 2 | 3 | 3 | 2 | 2 | 3 | **4** | 5 |
| A2. New public sector loans on less favorable terms in 2014-2034 2 | 2 | 3 | 4 | 4 | 5 | 5 | **9** | 12 |
| A3. Oil sector development starting from FY2013/14 | 2 | 3 | 5 | 5 | 5 | 6 | **4** | -2 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2015-2016 | 2 | 3 | 5 | 4 | 5 | 6 | **8** | 10 |
| B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/ | 2 | 3 | 4 | 4 | 4 | 5 | **8** | 9 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016 | 2 | 3 | 5 | 4 | 5 | 6 | **9** | 10 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/ | 2 | 3 | 5 | 5 | 5 | 6 | **9** | 9 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 2 | 3 | 4 | 3 | 4 | 4 | **7** | 9 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/ | 2 | 5 | 6 | 6 | 7 | 8 | **12** | 13 |
| *Memorandum item:* | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 16 | 16 | 16 | 16 | 16 | 16 | **16** | 16 |

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

## Table 2b. Uganda: Sensitivity Analysis for Key Indicators of Public Debt, 2014–2034

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2024 | 2034 |
| **PV of Debt-to-GDP Ratio** | | | | | | | | |
| **Baseline** | 27 | 28 | 31 | 31 | 30 | 30 | 27 | 23 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 27 | 26 | 26 | 25 | 24 | 24 | 22 | 20 |
| A2. Primary balance is unchanged from 2014 | 27 | 31 | 35 | 38 | 42 | 45 | 58 | 77 |
| A3. Permanently lower GDP growth 1/ | 27 | 28 | 31 | 31 | 31 | 31 | 31 | 35 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016 | 27 | 29 | 33 | 33 | 33 | 33 | 32 | 30 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-2016 | 27 | 28 | 29 | 29 | 29 | 28 | 26 | 23 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 27 | 27 | 28 | 28 | 28 | 28 | 27 | 25 |
| B4. One-time 30 percent real depreciation in 2015 | 27 | 32 | 35 | 34 | 33 | 33 | 30 | 28 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 27 | 38 | 40 | 40 | 39 | 38 | 33 | 27 |
| **PV of Debt-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 176 | 178 | 195 | 191 | 184 | 182 | 166 | 141 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 176 | 164 | 163 | 156 | 150 | 149 | 134 | 120 |
| A2. Primary balance is unchanged from 2014 | 176 | 194 | 221 | 239 | 257 | 274 | 356 | 468 |
| A3. Permanently lower GDP growth 1/ | 176 | 179 | 198 | 195 | 191 | 191 | 191 | 216 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016 | 176 | 182 | 206 | 204 | 200 | 200 | 195 | 185 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-2016 | 176 | 176 | 186 | 182 | 176 | 174 | 160 | 138 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 176 | 171 | 177 | 175 | 171 | 171 | 164 | 152 |
| B4. One-time 30 percent real depreciation in 2015 | 176 | 205 | 219 | 213 | 205 | 199 | 185 | 170 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 176 | 239 | 254 | 247 | 238 | 231 | 204 | 162 |
| **Debt Service-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 35 | 33 | 35 | 33 | 31 | 32 | 27 | 26 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 35 | 33 | 35 | 31 | 29 | 30 | 25 | 24 |
| A2. Primary balance is unchanged from 2014 | 35 | 33 | 36 | 34 | 33 | 35 | 37 | 50 |
| A3. Permanently lower GDP growth 1/ | 35 | 33 | 36 | 33 | 31 | 33 | 29 | 32 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2015-2016 | 35 | 34 | 37 | 34 | 32 | 33 | 29 | 29 |
| B2. Primary balance is at historical average minus one standard deviations in 2015-2016 | 35 | 33 | 35 | 33 | 30 | 32 | 27 | 25 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 35 | 33 | 35 | 33 | 30 | 32 | 27 | 26 |
| B4. One-time 30 percent real depreciation in 2015 | 35 | 34 | 37 | 35 | 33 | 35 | 32 | 33 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2015 | 35 | 33 | 38 | 36 | 33 | 34 | 31 | 27 |

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.